

Adversarial Robustness of CNN vs Transformer Architectures for Traffic Sign Recognition: A Comparative Study with Defense Evaluation

Awn Nashat Rihani

*Department of Computer Science
Jordan University of Science and Technology
Irbid, Jordan
anrihani25@cit.just.edu.jo*

Mohammad Rawashdeh

*Department of Computer Science
Jordan University of Science and Technology
Irbid, Jordan
mzrawshdeh256@cit.just.edu.jo*

Omar Darwish

*Information Security and Applied Computing
Eastern Michigan University
Ypsilanti, USA
odarwish@emich.edu*

Ali K. Jaber

*Technical College of Petroleum and Energy Engineering
Al-Awsat Technical University
Kufa, Iraq
alikadhemjabe.jaber@wmich.edu*

Abstract—Deep learning models are crucial in traffic sign recognition systems in autonomous driving, yet they are susceptible to adversarial perturbations, which poses a critical safety risk as a single misclassification can lead to dangerous consequences. However, adversarial attacks remain a serious and underexplored threat. Existing studies have evaluated adversarial robustness primarily on CNN-based architectures such as YOLOv8, overlooking the growing adoption of transformer-based models in computer vision. To bridge this gap, we conduct a comprehensive adversarial robustness comparison between YOLOv8 and Vision Transformers (ViT-B/16) on the German Traffic Sign Recognition Benchmark (GTSRB). Beyond evaluating vulnerability we systematically assess two defense methods, median blur preprocessing and adversarial training, where the training data is progressively enriched with attacks of increasing strength, using FGSM alone, then using FGSM combined with PGD, and finally using C&W attacks with FGSM and PGD, providing insights into which architecture is more defensible under real-world threat conditions. Our results reveal that ViT-B/16 demonstrates significantly better robustness under iterative attacks and responds more effectively to adversarial training, while YOLOv8 shows resilience under optimization-based attacks (C&W). These findings establish a critical foundation for designing more robust traffic sign recognition systems in autonomous driving.

Index Terms—adversarial robustness, traffic sign recognition, YOLOv8, Vision Transformer, FGSM, PGD, Carlini-Wagner, adversarial training, autonomous driving, GTSRB

I. INTRODUCTION

Detecting and identifying traffic signs in autonomous driving systems (ADS) is a critical task, where any misclassification can make the system malfunction or make hazardous decisions, including driving at improper speeds, failing to stop or yield, or navigating into restricted zones [1], [2], leading to catastrophic danger for passengers and bystanders. Deep learning models are now the dominant tool for computer vision. In life-critical contexts like autonomous driving, classifiers must

do more than perform well on clean data — they must also stay robust against real-world noise and adversarial attacks, which deliberately craft perturbations that induce confident misclassifications [3], [4].

The systematic study of adversarial vulnerability in deep neural networks was first studied by Goodfellow et al. [3], who introduced the Fast Gradient Sign Method (FGSM), a computationally efficient single-step attack that exploits the local linearity of deep networks to maximize classification loss. Madry et al. [5] later made adversarial robustness a min-max optimization problem, proposing Projected Gradient Descent (PGD) as the benchmark for robustness evaluation. Carlini and Wagner [6] further advanced the field by introducing an optimization-based attack that minimizes perturbation distortion, circumventing purported defenses; it is widely considered as the most powerful white box adversarial attack. Now, FGSM, PGD and C&W are the standard white box attacks and threats against which classifiers are evaluated [7], [8].

To address the adversarial attacks problem there are two main strategies used for defense in literature. These include input-level preprocessing methods, such as median blurring, which limit and suppress adversarial noise before inference without retraining the model, offering a lightweight defense; and adversarial training, which augments the training data with adversarial examples to enforce perturbation-invariant representations [5]. Zhang et al. [9] refined this framework through establishing a trade-off between natural accuracy and robustness using the TRADES objective. Despite their effectiveness, these defenses have mainly been evaluated on convolutional network architectures, leaving their behavior on transformer-based models underexplored [10], [11].

The introduction of Vision Transformers (ViT) by Dosovits-

skiy et al. [12] changed convolutional inductive bias, replacing local feature extraction with global self-attention mechanisms and achieving competitive performances across standard benchmarks. This change in architecture raised questions regarding adversarial robustness. Bai et al. [13] argued that ViTs and CNNs show comparable vulnerability under fair evaluation conditions, while Ali et al. [14] showed that CNNs display greater resilience to simple gradient-based attacks such as FGSM, but ViTs show a stronger resistance against more powerful iterative attacks such as PGD. A recent survey by Fawole and Rawat [15] specifically identified the comparative robustness of ViTs and CNNs in TSR as a question that needs investigation. Meanwhile, YOLOv8 [16] has emerged as the dominant CNN-based architecture in real-world ADS perception pipelines, yet its adversarial robustness relative to transformer-based alternatives remains uncharacterized under systematic evaluation [4], [8].

Despite this body of work, a critical gap remains in the literature. Chen et al. [1], the most closely related study, systematically evaluated adversarial attacks and defenses in ADS perception tasks exclusively through the lens of YOLOv8, without considering transformer-based architectures. Wang et al. [2] extended the evaluation to commercial TSR systems under physical-world attacks, yet did not include architectural comparisons between CNNs and ViTs. No prior work has conducted a controlled head-to-head empirical comparison of YOLOv8 and ViT-B/16 under identical white-box attack conditions and defense configurations on the German Traffic Sign Recognition Benchmark (GTSRB) [17].

This paper aims to address this gap and present the first systematic comparative study of adversarial robustness between YOLOv8n-cls and ViT-B/16 on GTSRB [17] under three main white-box attacks: FGSM [3], PGD [5], and C&W [6] and two complementary defense strategies: median blur preprocessing and TRADES-based adversarial training [9] across three progressively augmented training configurations. Our principal contributions are as follows:

- We demonstrate a significant architectural disparity under PGD attack, where YOLOv8 collapses to 9.2% accuracy while ViT-B/16 maintains 47.5% under identical attack conditions with no defense, representing a 38 percentage point gap that persists across all adversarial training configurations [5].
- We show that ViT-B/16 preserves clean accuracy under adversarial training, degrading marginally from 97.0% to 95.9% under the full mixed training configuration, whereas YOLOv8 suffers a consistent clean accuracy drop from 92.6% to 88.7%, revealing a fundamental architectural difference in the robustness-accuracy trade-off [9].
- We establish that YOLOv8 shows a better resilience under C&W attack at baseline, achieving 41.3% compared to ViT-B/16's 29.5%, suggesting that optimization-based attacks damage transformer architectures more effectively than gradient-based iterative attacks [6].
- We demonstrate that median blur preprocessing dispro-

portionately benefits YOLOv8 under PGD, recovering accuracy from 9.2% to 35.7%, a gain of 26.5 percentage points, while offering only a small improvement to ViT-B/16 from 47.5% to 50.4%, indicating that preprocessing defenses are architecture-dependent in their effectiveness [1].

- We show that architecture choice is a more decisive factor than defense strategy for PGD robustness as the undefended ViT-B/16 baseline at 47.5% outperforms the best adversarially trained YOLOv8 configuration at 31.6%, suggesting that model selection should precede defense design in safety-critical ADS deployments [13], [14].

These findings provide actionable, architecture-aware guidance for the design of robust perception systems in safety-critical autonomous driving applications.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset, model architectures, attack configurations, and defense strategies employed. Section IV presents and analyzes the experimental results. Section V discusses the key findings and their implications for autonomous driving systems, and Section VI concludes the paper.

II. RELATED WORK

A. Adversarial Attacks on Deep Neural Networks

Adversarial attacks pose a fundamental threat to the reliable deployment of deep learning models in safety-critical systems. Since small perturbation attacks like FGSM can reliably induce misclassification in state-of-the-art classifiers [3], the field has rapidly evolved toward more powerful attacks. Iterative optimization-based methods such as PGD [5] and C&W [6] have demonstrated that single-step attacks represent only a lower bound on model vulnerability, with stronger iterative adversaries exposing deeper weaknesses. The severity of these vulnerabilities across computer vision applications have been documented [4], [7], [8], [18], with consistent findings that no architecture class has demonstrated immunity to well-crafted white-box adversarial inputs.

B. Adversarial Robustness of Vision Transformers

The introduction of Vision Transformers [12] prompted significant interest in whether the self-attention mechanism has better robustness advantages over convolutional architectures. Shao et al. [19] demonstrated that pure ViTs achieve higher robust accuracy than CNNs, attributing this to the tendency of ViTs to learn low-frequency features that are less affected by adversarial noise. Benz et al. [20] reached similar results under C&W and DDN attacks, confirming that CNN based architecture is the least robust architecture in multiple settings. Bai et al. [13] suggests that the robustness advantage of ViTs over CNNs diminishes considerably when model capacity and training conditions are properly controlled. Ali et al. [14] further refined this picture by demonstrating that the relationship depends on the attack type as CNNs show greater resilience against simple gradient-based attacks such as FGSM, while ViTs perform better than CNNs against stronger

iterative attacks like PGD. Jain and Dutta [21] reported that softmax operations in attention blocks cause gradient masking that weakens standard attacks like PGD and C&W, meaning ViT robustness may be partially inflated in previous evaluations. Collectively, these findings establish that the robustness relationship between ViTs and CNNs is both attack-dependent and architecture-specific, motivating the need for controlled comparisons on domain-specific benchmarks.

C. Adversarial Defenses

Adversarial training remains the most widely adopted and accepted defense against adversarial attacks. Madry et al. [5] established its foundations by training models on PGD-generated adversarial examples, demonstrating huge robustness improvement at the cost of some clean accuracy. Zhang et al. [9] advanced this approach through the TRADES objective, which formalizes the robustness-accuracy trade-off and has since become the standard framework for robust training. Zhao et al. [22] surveyed the evolution of adversarial training techniques and found that robustness gains are heavily dependent on both model architecture and the specific attacks used during training. Shao et al. [19] found that adversarial training transfers effectively to ViTs, while Jain and Dutta [21] showed that using stronger, more accurate attacks during training leads to more robust ViT models. Complementing training-based approaches, input preprocessing methods such as median blurring offer a lightweight defence that requires no model modification or retraining, operating by smoothing adversarial perturbations before inference [1]. While effective against weaker attacks, Chen et al. [1] found that its benefits are limited against stronger iterative methods, suggesting that preprocessing alone is insufficient as a standalone defense.

D. Adversarial Attacks on Traffic Sign Recognition

Traffic sign recognition has become an important domain for adversarial robustness research given its central role in autonomous driving safety. Stallkamp et al. [17] established GTSRB as the standard benchmark for traffic sign classification, providing a realistic multi-class evaluation setting that has been widely adopted in robustness studies. Pavlitska et al. [23] surveyed adversarial attacks on TSR models across both digital and physical attack settings. Chen et al. [1] systematically evaluated adversarial attacks and defenses on ADS perception tasks including traffic sign classification using YOLOv8, demonstrating that adversarial training and preprocessing defenses exhibit attack-dependent effectiveness. Wang et al. [2] extended this evaluation to commercial TSR systems under physical-world adversarial patch attacks, revealing that academic attack success rates do not generalize reliably to real-world deployments. Badjie et al. [24] provided a systematic review of adversarial attacks in ADS image classification, reinforcing the need for architecture-aware defense strategies. Fawole and Rawat [15] specifically identified the comparative adversarial robustness of ViTs and CNNs in traffic sign recognition as an open empirical question, noting the absence of controlled comparisons under unified experimental protocols.

This paper aims to address this gap by providing the first systematic comparison of YOLOv8 and ViT-B/16 on GTSRB under identical white-box attack and defense conditions.

III. METHODOLOGY

A. Dataset

All experiments were conducted on the German Traffic Sign Recognition Benchmark (GTSRB) [17], a widely used benchmark for traffic sign classification comprising 43 classes and 51,839 real-world traffic sign images with varying light, occlusion, and perspectives. The original training set of 39,209 images was partitioned into a training set of 31,367 images and a validation set of 7,842 images using a stratified 80/20 random split for the validation set. The test set consisted of 12,630 images with ground truth labels and was used exclusively for final evaluation. All images were resized to 64×64 pixels and normalized with GTSRB-specific channel statistics: mean = $[0.3337, 0.3064, 0.3171]$ and standard deviation = $[0.2672, 0.2564, 0.2629]$.

B. Model Architectures

1) *YOLOv8*: We employ YOLOv8n-cls, the nano classification variant of the YOLOv8 architecture [16], using the Ultralytics framework v8.4.46. The model was configured for 43-class classification with an input size of 64×64 pixels and trained for 20 epochs with a batch size of 64 using the AdamW optimizer with automatically tuned hyperparameters.

2) *ViT-B/16*: We employ ViT-B/16 [12], loaded via the timm library using `vit_base_patch16_224` with ImageNet pretraining. The model was adapted to the 64×64 input resolution through timm's built-in positional embedding interpolation, and the classification head was replaced with a 43-class output layer. The model was fine-tuned on GTSRB with no frozen layers, including the patch embedding, all 12 transformer blocks, and the classification head. Training was conducted for 20 epochs with a batch size of 64, using the Adam optimizer with a learning rate of 10^{-4} and weight decay of 10^{-4} . A CosineAnnealingLR scheduler with $T_{\max} = 20$ was applied to gradually reduce the learning rate throughout training.

C. Adversarial Attacks

All attacks are white-box evasion attacks applied at inference time, with full access to model parameters and gradients. No poisoning or physical-world attacks are considered. Adversarial examples are generated using the torchattacks library. We evaluate three main attacks:

1) *Fast Gradient Sign Method (FGSM)*: FGSM [3] generates adversarial examples in a single gradient step by perturbing the input in the direction that maximizes the classification loss:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y)) \quad (1)$$

We use a perturbation of $\epsilon = 8/255$ under the ℓ_∞ -norm.

2) *Projected Gradient Descent (PGD)*: PGD [5] extends FGSM to an iterative multi-step attack, applying multiple small gradient steps while projecting the perturbation back onto the ℓ_∞ -ball of radius ϵ after each step:

$$x_{adv}^{t+1} = \Pi_{x+S} (x_{adv}^t + \alpha \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x_{adv}^t, y))) \quad (2)$$

We use $\epsilon = 8/255$, step size $\alpha = 2/255$, and 10 iterations.

3) *Carlini & Wagner (C&W)*: The C&W attack [6] frames the attack as a constrained optimization problem, minimizing the ℓ_2 perturbation norm while ensuring misclassification:

$$\min_{\delta} \|\delta\|_2 + c \cdot f(x + \delta) \quad (3)$$

where f is an objective function that is negative if and only if the attack succeeds. We use confidence $\kappa = 0$, regularization constant $c = 1$, 50 optimization steps, and a learning rate of 0.01.

D. Defense Strategies

1) *Median Blur Preprocessing*: Median blur is applied as an input-level preprocessing defense by replacing each pixel value with the median of its local neighborhood prior to classification. We use a kernel size of 3, which smooths high-frequency adversarial perturbations while preserving the semantic content of the traffic sign images. This defense does not require retraining and is applied directly to adversarial test images at inference time [1].

2) *Adversarial Training*: Adversarial training augments the training distribution with adversarial examples to force models to learn perturbation-invariant representations [5]. All adversarial training configurations follow the TRADES framework [9]. The clean training set of 31,367 images is always included in all training iterations, and adversarial examples are added incrementally as sets of 31,367 adversarial images per attack type. Adversarial validation is performed every epoch using both clean and PGD-attacked versions of the validation set, and the best model checkpoint is saved based on clean validation accuracy. We evaluate three progressively augmented configurations:

- **Adv(FGSM)**: 31,367 clean images + 31,367 FGSM examples = 62,734 total training samples.
- **Adv(FGSM+PGD)**: 31,367 clean images + 31,367 FGSM examples + 31,367 PGD examples = 94,101 total training samples.
- **Adv(FGSM+PGD+C&W)**: 31,367 clean images + 31,367 FGSM examples + 31,367 PGD examples + 31,367 C&W examples = 125,468 total training samples.

E. Evaluation

Each model configuration is evaluated on the official GT-SRB test set of 12,630 images under seven conditions: clean (no attack), FGSM, FGSM with median blur, PGD, PGD with median blur, C&W, and C&W with median blur. This yields a total of 56 evaluation settings across the two architectures and four training configurations (baseline plus three adversarial training variants). All adversarial examples are generated at

inference time using white-box access to the model. Performance is reported as classification accuracy on the full test set.

IV. RESULTS

A. Baseline Performance

On the clean test set, YOLOv8 achieves 92.6% accuracy while ViT-B/16 achieves 97.0%, demonstrating that the transformer-based model produces stronger baseline conditions. This result is consistent with studies showing that ViTs achieve a better performance over CNNs [12], [19].

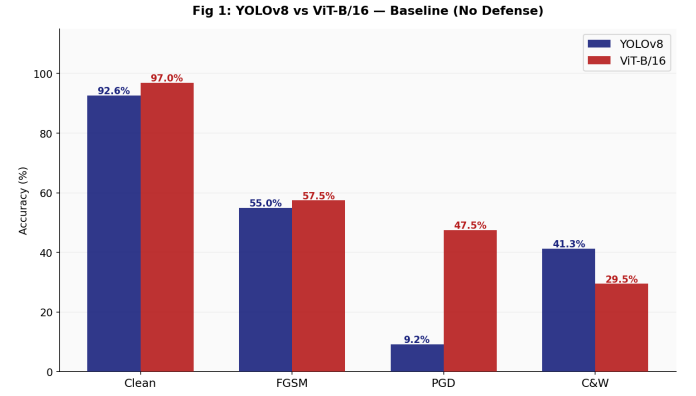


Fig. 1. Baseline classification accuracy of YOLOv8 and ViT-B/16 on clean and adversarial inputs (FGSM, PGD, C&W) with no defense applied.

B. Performance Under Adversarial Attacks

1) *FGSM*: Under FGSM attack, both models experience a significant accuracy drop. YOLOv8 falls from 92.6% to 55.0%, while ViT-B/16 falls from 97.0% to 57.5%. The performance of the two architectures is similar in this single-step attack, suggesting that they are both vulnerable to gradient-based perturbations at this level. This finding is consistent with Ali et al. [14], who also observed that CNNs and ViTs show comparable performance, and in some cases CNNs had better robustness under simple gradient-based attacks. Applying median blur offers very little improvement for both models, with YOLOv8 dropping slightly to 54.0% and ViT-B/16 to 57.3%, indicating that FGSM perturbations are not effectively mitigated by local smoothing, a result that aligns with observations by Chen et al. [1].

2) *PGD*: PGD reveals a striking architectural difference in this study. YOLOv8 collapses to 9.2% accuracy, near random-chance performance across 43 classes, while ViT-B/16 retains 47.5% accuracy, a gap of 38.3 percentage points. This result aligns with the findings of Shao et al. [19] and Benz et al. [20], who found that ViTs maintain a higher robust accuracy than CNNs under iterative attacks like PGD, attributing this advantage to the tendency of ViTs to rely on low-frequency features that are less affected by adversarial perturbations. Ali et al. [14] also found that ViTs hold a robustness advantage specifically under powerful iterative attacks, consistent with

TABLE I
CLASSIFICATION ACCURACY (%) OF YOLOV8 AND ViT-B/16 ON GTSRB UNDER ADVERSARIAL ATTACKS AND DEFENSES

Model	Training Config	Clean	FGSM	FGSM+Blur	PGD	PGD+Blur	C&W	C&W+Blur
YOLOv8	Baseline	92.6	55.0	54.0	9.2	35.7	41.3	45.8
YOLOv8	Adv(FGSM)	88.7	63.2	64.9	24.6	51.7	56.6	50.6
YOLOv8	Adv(FGSM+PGD)	88.9	64.3	65.5	27.8	53.7	58.2	49.7
YOLOv8	Adv(FGSM+PGD+C&W)	89.5	64.9	65.8	31.6	54.3	59.1	49.8
ViT-B/16	Baseline	97.0	57.5	57.3	47.5	50.4	29.5	42.0
ViT-B/16	Adv(FGSM)	97.1	66.7	67.0	56.5	59.6	32.0	53.5
ViT-B/16	Adv(FGSM+PGD)	96.8	65.4	65.9	56.2	59.5	33.6	52.0
ViT-B/16	Adv(FGSM+PGD+C&W)	95.9	65.0	65.6	55.9	59.2	33.7	51.1

our observations on GTSRB. Median blur provides substantial recovery for YOLOv8, raising accuracy from 9.2% to 35.7%, while offering only marginal improvement for ViT-B/16 (47.5% to 50.4%). The disproportionate benefit of pre-processing for YOLOv8 under PGD is also similar to the findings of Chen et al. [1].

3) C&W: Under the optimization-based C&W attack, the robustness pattern reverses. YOLOv8 achieves 41.3% accuracy while ViT-B/16 drops to only 29.5%, suggesting that the transformer architecture is more susceptible to minimal-perturbation attacks. Jain and Dutta [21] showed that the use of softmax functions within the ViT attention layers leads to gradient masking, which makes standard attacks ineffective, thus, the C&W attack, which explicitly deals with gradient masking, is particularly effective against ViT architectures. The median blur filter helped improve the performance of both networks slightly, achieving 45.8% for YOLOv8 and 42.0% for ViT-B/16.

C. Effect of Adversarial Training

1) *FGSM Attack*: Adversarial training improves FGSM robustness for both architectures. ViT-B/16 trained on FGSM examples achieves 66.7% under FGSM, an improvement of 9.2 percentage points over baseline. YOLOv8 improves from 55.0% to 63.2%. Adding PGD and C&W examples to the training mix provides only marginal additional gains for both models, suggesting diminishing returns from more diverse adversarial training against FGSM.

2) *PGD Attack*: Adversarial training substantially improves PGD robustness for ViT-B/16, with Adv(FGSM) raising accuracy from 47.5% to 56.5%, and the full mix configuration reaching 55.9%. However, YOLOv8's PGD robustness is still low across all adversarial training configurations, reaching only 31.6% even under full mix training. This confirms that the architectural vulnerability of YOLOv8 to iterative attacks cannot be fully solved or improved with adversarial training alone, showing similar results to Shao et al. [19] that architectural bias plays a main role in determining robustness, independent of the training strategy employed.

3) *C&W Attack*: The most notable improvement of robustness from adversarial training is observed under C&W for YOLOv8. Full mix training raises YOLOv8 accuracy under C&W from 41.3% to 59.1%, a gain of 17.8 percentage points.

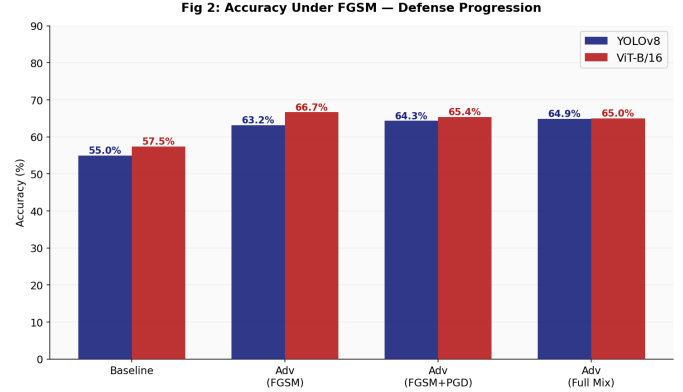


Fig. 2. Accuracy under FGSM attack across four training configurations: Baseline, Adv(FGSM), Adv(FGSM+PGD), and Adv(FGSM+PGD+C&W). Both models improve under adversarial training, with diminishing returns as training diversity increases.

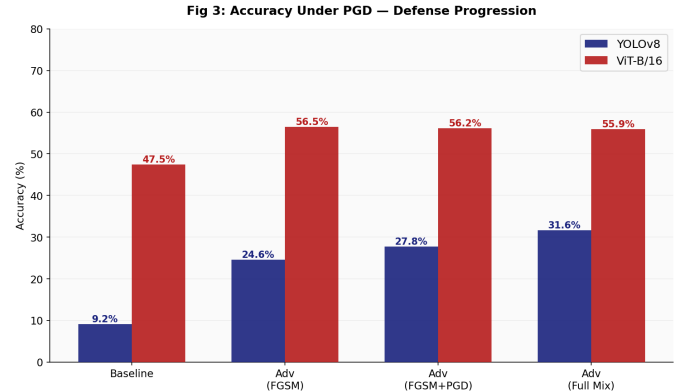


Fig. 3. Accuracy under PGD attack across four training configurations.

However, ViT-B/16 shows little improvement under C&W across all adversarial training configurations, reaching only 33.7% under full mix training. This asymmetry suggests that CNN-based architectures benefit more from diverse adversarial training against optimization-based attacks than transformer-based models, a finding consistent with Zhao et al. [22].

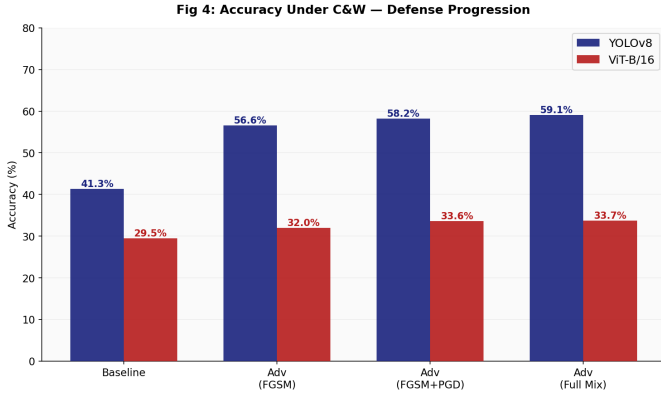


Fig. 4. Accuracy under C&W attack across four training configurations.

D. Clean Accuracy Tradeoff

An accuracy-robustness tradeoff is observed across both architectures. YOLOv8 experiences a drop in clean accuracy from 92.6% to 88.7% when trained with FGSM adversarial examples, with only minimal recovery across more diverse configurations. ViT-B/16, in contrast, maintains clean accuracy at 97.1% under Adv(FGSM) before experiencing a modest drop to 95.9% under full mix training. This indicates that ViT-B/16 is better at preserving clean accuracy while gaining adversarial robustness, consistent with the TRADES framework [9] and with empirical findings by Shao et al. [19].

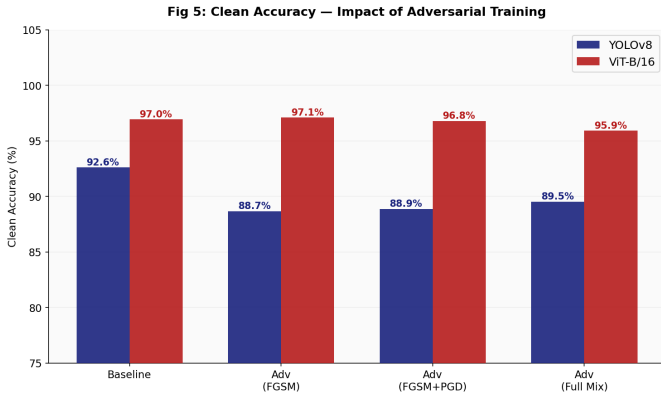


Fig. 5. Clean test accuracy across four training configurations. ViT-B/16 preserves clean accuracy above 95.9% across all settings, while YOLOv8 drops to 88.7% under Adv(FGSM) training and does not recover under more diverse configurations.

E. Combined Defense: Adversarial Training and Median Blur

Combining adversarial training with median blur preprocessing yields the strongest overall defense for both models. Under PGD, YOLOv8 trained with the full mix and evaluated with median blur achieves 54.3%, compared to 9.2% at baseline with no defense — a recovery of 45.1 percentage points. ViT-B/16 under the same conditions achieves 59.2%. Under C&W with median blur, YOLOv8 full mix reaches 49.8% while ViT-B/16 reaches 51.1%, suggesting that the combined

defense partially closes the architectural gap between the two models under optimization-based attacks.

F. Comparison with Prior Work

Direct numerical comparison with prior studies is complicated by differences in evaluation metrics, datasets, and attack configurations. Chen et al. [1] evaluated YOLOv8 on a traffic sign detection task using mAP@50 rather than classification accuracy, and reported that median blurring improved detection performance substantially under Gaussian noise and FGSM but offered limited gains under Auto-PGD. Our classification results on GTSRB are qualitatively consistent with this pattern. Bai et al. [13] and Ali et al. [14] both evaluated ViT vs CNN robustness on general benchmarks such as CIFAR-10 and ImageNet, finding that the robustness advantage of ViTs is attack-dependent. Our results on GTSRB corroborate and extend these findings to the traffic sign recognition domain, demonstrating that the same attack-dependent pattern holds in a safety-critical real-world setting: ViTs hold a decisive advantage under PGD while CNNs show greater baseline resilience under C&W.

TABLE II
QUALITATIVE COMPARISON WITH PRIOR WORK

Study	Model	Dataset	Metric	Key Finding
Chen et al. [1]	YOLOv8	TSR	mAP@50	Blur aids FGSM, not PGD
Ali et al. [14]	ViT vs CNN	CIFAR-10, ImageNet	Accuracy	CNN best FGSM, ViT best PGD
Bai et al. [13]	ViT vs CNN	ImageNet	Accuracy	Gap closes with aligned training
Shao et al. [19]	ViT vs CNN	ImageNet	Rob. Acc.	ViTs more robust under PGD
Ours	YOLOv8 vs ViT-B/16	GTSRB	Accuracy	ViT 47.5% vs YOLO 9.2% PGD

V. DISCUSSION

A. Interpretation of Key Findings

Our results reveal a fundamental architectural difference in adversarial robustness that plays out differently depending on the type of attack. Under PGD, the iterative nature of the attack takes advantage of the local feature representations learned by YOLOv8’s convolutional layers, producing perturbations that align with the high-frequency patterns the model depends on for classification. ViT-B/16, in contrast, spreads its representations across global self-attention patches that focus on low-frequency structural information, which makes it naturally more resistant to this kind of perturbation. This explanation is backed by Shao et al. [19], who showed that ViTs learn far fewer high-frequency features than CNNs and that this property translates directly into stronger adversarial robustness. The 38.3 percentage point gap we see under PGD on GTSRB fits this explanation when applied to the traffic sign recognition domain.

The reverse pattern under C&W — where YOLOv8 beats ViT-B/16 by 11.8 percentage points at baseline — is more

Fig 6: Effect of Median Blur Defense Across All Configurations

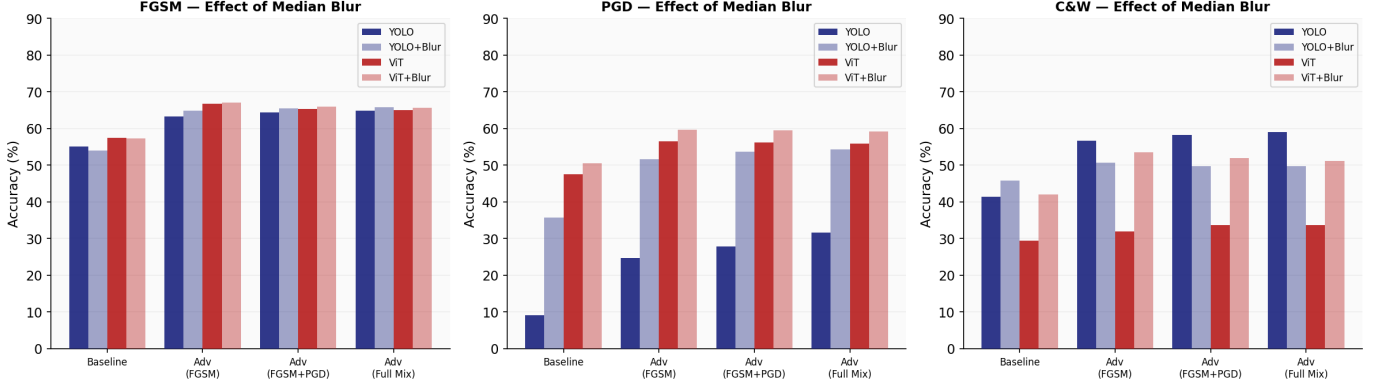


Fig. 6. Effect of median blur preprocessing across FGSM, PGD, and C&W attacks for all training configurations. Blur offers negligible benefit under FGSM, substantial recovery for YOLOv8 under PGD, and moderate improvement for both models under C&W.

Fig 7: Accuracy Drop Under Attack — All Configurations (lower is better)

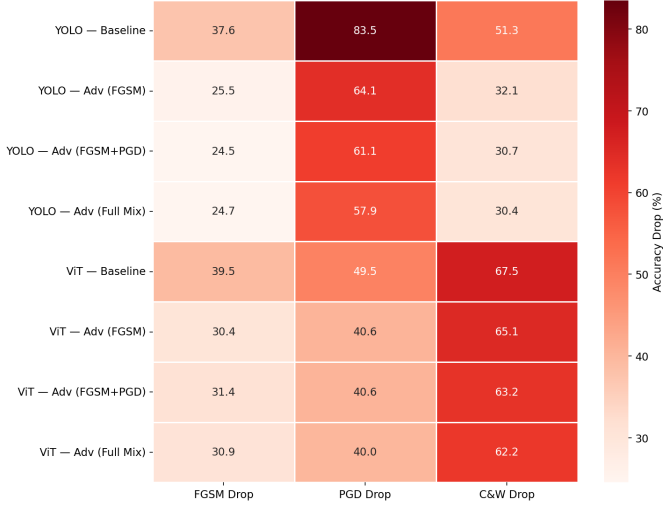


Fig. 7. Heatmap of accuracy degradation across all model and training configurations under each attack. Darker red indicates greater accuracy loss relative to clean baseline. YOLOv8 baseline shows the most severe PGD degradation, while ViT-B/16 baseline shows the most severe C&W degradation.

subtle. Jain and Dutta [21] found that softmax operations in ViT attention blocks cause floating-point underflow, which masks gradients and weakens standard attacks. Because C&W minimizes perturbation distortion through an ℓ_2 optimization framework rather than relying purely on softmax gradients, it is well-positioned to bypass this masking effect. Our results suggest the same mechanism is at play on GTSRB: C&W exploits a vulnerability in ViT-B/16 that neither PGD nor FGSM can reach, while YOLOv8’s CNN structure, which does not suffer from softmax-induced masking, shows no comparable optimization-based weakness.

B. Implications for Autonomous Driving Systems

These findings have direct implications for the design of robust traffic sign recognition systems in safety-critical con-

texts. First, architectural choice should come before defense selection: the undefended ViT-B/16 baseline outperforms even the most heavily adversarially trained YOLOv8 configuration under PGD, suggesting that no amount of training-time augmentation can fully compensate for an architecturally vulnerable backbone. Second, the defense strategy needs to match the expected threat. For deployments mainly concerned with iterative gradient-based attacks, ViT-B/16 combined with adversarial training offers the strongest robustness profile. For deployments worried about optimization-based attacks such as C&W, YOLOv8 with full mix adversarial training provides much better resilience than ViT-B/16 under the same conditions. Third, ViT-B/16 keeps its clean accuracy nearly intact across all adversarial training configurations, which makes it the more practical choice when both clean performance and robustness matter — a common requirement in production ADS, where false negatives on benign inputs are just as unacceptable as misclassifications under attack.

The combined defense of adversarial training and median blur preprocessing gave the best overall robustness for both architectures, and notably narrowed the gap between YOLOv8 and ViT-B/16 under PGD (54.3% vs 59.2%) and C&W (49.8% vs 51.1%). This suggests that in real-world ADS, layered defenses that combine training-time and inference-time mitigations are a better bet than relying on either approach alone.

C. Limitations and Future Work

A few limitations of our study should be noted. First, our comparison uses standard deployment configurations for each architecture, which differ in optimizer, learning rate scheduling, and augmentation strategies. Bai et al. [13] have shown that controlled architectural comparisons with fully aligned training recipes can narrow the robustness gap between CNNs and ViTs; controlled comparisons of this kind in the traffic sign recognition setting are left to future work. Second, we restrict our evaluation to white-box evasion attacks in the digital domain. Physical-world attacks using printed adversarial patches or projected perturbations [2] represent a separate

and complementary threat model that needs its own evaluation. Third, our defenses are limited to median blur preprocessing and TRADES-based adversarial training; certified defenses, ensemble methods, and diffusion-based purification remain promising directions for future work. Finally, although GT-SRB is the standard benchmark for traffic sign classification, extending this comparison to detection tasks and to real-world driving datasets would strengthen how generally our findings apply.

VI. CONCLUSION

This paper presented the first systematic comparison of YOLOv8n-cls and ViT-B/16 on the German Traffic Sign Recognition Benchmark under three canonical white-box adversarial attacks (FGSM, PGD, and C&W) and two complementary defense strategies (median blur preprocessing and TRADES-based adversarial training across three progressively augmented configurations). Our findings demonstrate that adversarial robustness is strongly architecture-dependent and attack-dependent. ViT-B/16 exhibits substantially greater resilience under PGD than YOLOv8, with a 38.3 percentage point gap at baseline that persists across all adversarial training configurations, while YOLOv8 shows comparatively stronger robustness under C&W. ViT-B/16 also preserves clean accuracy under adversarial training far more effectively than YOLOv8, reinforcing its suitability for deployments that require both high clean performance and adversarial resilience. We further showed that median blur preprocessing disproportionately benefits YOLOv8 under PGD, and that combining adversarial training with median blur produces the strongest overall defense for both architectures, partially closing the architectural gap. These results provide actionable, architecture-aware guidance for the design of robust perception systems in safety-critical autonomous driving applications and underscore that architectural choice should be treated as a first-class design decision alongside defense strategy. Future work will extend this comparison to controlled training-recipe-aligned evaluations, physical-world attack scenarios, and broader sets of defenses including certified and diffusion-based methods.

REFERENCES

- [1] C. Chen, Y. Wang, N. S. Munir, X. Zhou, and X. Zhou, "Revisiting adversarial perception attacks and defense methods on autonomous driving systems," arXiv preprint arXiv:2505.11532, 2025.
- [2] N. Wang, S. Xie, T. Sato, Y. Luo, K. Xu, and Q. A. Chen, "Revisiting physical-world adversarial attack on traffic sign recognition: A commercial systems perspective," in *Proc. NDSS*, 2025.
- [3] I. J. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and harnessing adversarial examples," in *Proc. ICLR*, 2015.
- [4] X. Yuan, P. He, Q. Zhu, and X. Li, "Adversarial examples: Attacks and defenses for deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2805–2824, 2019.
- [5] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. ICLR*, 2018.
- [6] N. Carlini and D. Wagner, "Towards evaluating the robustness of neural networks," in *Proc. IEEE S&P*, 2017, pp. 39–57.
- [7] N. Akhtar, A. Mian, N. Kardan, and M. Shah, "Advances in adversarial attacks and defenses in computer vision: A survey," *IEEE Access*, vol. 9, pp. 155161–155196, 2021.
- [8] J. C. Costa, T. Roxo, H. Proença, and P. R. M. Inácio, "How deep learning sees the world: A survey on adversarial attacks & defenses," *IEEE Access*, vol. 12, pp. 61113–61136, 2024.
- [9] H. Zhang, Y. Yu, J. Jiao, E. Xing, L. El Ghaoui, and M. Jordan, "Theoretically principled trade-off between robustness and accuracy," in *Proc. ICML*, 2019, pp. 7472–7482.
- [10] N. Akhtar and A. Mian, "Threat of adversarial attacks on deep learning in computer vision: A survey," *IEEE Access*, vol. 6, pp. 14410–14430, 2018.
- [11] K. Xu et al., "A survey of adversarial examples in computer vision: Attack, defense, and beyond," *IEEE Trans. Circuits Syst. Video Technol.*, 2024.
- [12] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.
- [13] Y. Bai, J. Mei, A. L. Yuille, and C. Xie, "Are transformers more robust than CNNs?" in *Proc. NeurIPS*, vol. 34, 2021, pp. 26831–26843.
- [14] K. Ali et al., "Adversarial robustness of vision transformers versus convolutional neural networks," *IEEE Access*, vol. 12, pp. 105281–105293, 2024.
- [15] O. Fawole and D. B. Rawat, "Recent advances in vision transformer robustness against adversarial attacks in traffic sign detection and recognition: A survey," *ACM Comput. Surv.*, vol. 57, no. 10, 2025.
- [16] Ultralytics, "YOLOv8," 2025. [Online]. Available: <https://docs.ultralytics.com/models/yolov8/>
- [17] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "Man vs. computer: Benchmarking machine learning algorithms for traffic sign recognition," *Neural Networks*, vol. 32, pp. 323–332, 2012.
- [18] Y. Li, B. Xie, S. Guo, Y. Yang, and B. Xiao, "A survey of robustness and safety of 2D and 3D deep learning models against adversarial attacks," *ACM Comput. Surv.*, vol. 56, no. 6, pp. 1–37, 2024.
- [19] R. Shao, Z. Shi, J. Yi, P. Chen, and C. Hsieh, "On the adversarial robustness of vision transformers," *Trans. Mach. Learn. Res.*, 2022.
- [20] P. Benz, S. Ham, C. Zhang, A. Karjauv, and I. S. Kweon, "Adversarial robustness comparison of vision transformer and MLP-Mixer to CNNs," in *Proc. ICCV Workshops*, 2021.
- [21] S. Jain and T. Dutta, "Towards understanding and improving adversarial robustness of vision transformers," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 24736–24745.
- [22] M. Zhao, L. Zhang, J. Ye, H. Lu, B. Yin, and X. Wang, "Adversarial training: A survey," arXiv preprint arXiv:2410.15042, 2024.
- [23] S. Pavlitska, N. Lambing, and J. M. Zöllner, "Adversarial attacks on traffic sign recognition: A survey," in *Proc. Int. Conf. Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*, 2023, pp. 1–6.
- [24] B. Badjie, J. Cecilio, and A. Casimiro, "Adversarial attacks and countermeasures on image classification-based deep learning models in autonomous driving systems: A systematic review," *ACM Comput. Surv.*, vol. 57, no. 1, pp. 1–52, 2024.